How Low Cost Carriers Withdrawal Affects Airfares in U.S. Domestic Markets

by

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ABSTRACT

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This paper examines how airfare prices fluctuate following the exit of a low-cost carrier (LCC) in a U.S. domestic market. Most of the current literature on airline economics entry and exit focuses on LCC entry; this paper addresses LCC exit. The paper uses the DB1B Market data from 2021 Q1 to 2023 Q4 from U.S. Department of Transportation, Bureau of Transportation Statistics (2023). Employing a difference-in-difference (DiD) model to compare the treated routes at Albany International Airport (ALB) to the control routes of Syracuse Hancock Airport (SYR). Based on the results of this model, the paper finds a 7.15% increase in prices for the treated routes. Through several robustness checks, such as a parallel trends test and a spill-over effect test, the paper assesses the validity of the empirical framework. These findings are consistent in magnitude with those of LCC entry and underscore the importance of LCC in market competition.

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CHAPTER ONE

Introduction

When consumers in the United States purchase an airline ticket, they generally face two options. One is to fly with a legacy carrier, such as Delta Air Lines, United Airlines, or American Airlines, which traditionally offer extensive route networks, multiple cabin classes, and premium services that consumers can elect to pay for. The second option is to book with an LCC, a low-cost business model represented by airlines like Southwest, Frontier, and Spirit that competes with legacy carriers primarily through lower fares and a basic offering of services, allowing firms to keep prices low. Since the deregulation of the U.S. airline industry in 1978, these two business models have thrived in an evolving competitive environment that has shaped both market structure and consumer behavior. The entry of LCCs was historically associated with large reductions in fares, greater consumer choices, and increased efficiency in the industry. Previous studies, such as Ito and Lee (2003), Goolsbee and Syverson (2008), and Morrison (2001), show that even the threat of low-cost entry could force downward pressure on ticket prices. These findings reflected an era in which deregulation and the expansion of LCCs reshaped the economic landscape of air travel, driving substantial gains in consumer welfare.

Over time, the industry has become increasingly concentrated following the mergers in the 2000s and 2010s. Through these mergers, the national market has become consolidated with a CR_4 of 68.6, as shown on the aviation dashboard of BTS. While the market has consolidated, firms have optimized their operations. Specifically look-

ing at the big three legacy carriers, adapting a fare group that matches the style of the LCC business model. However, with this, the legacy carriers are also able to retain their brand recognition and expansive frequent flier programs, as well as their ability to sell higher-class fares to willing to pay consumers. As Borenstein and Rose (2014) explain, the early benefits of deregulation, such as lower average fares and rapid passenger growth, eventually faded as fare dispersion grew and a few large airlines began to dominate the market. Today, this trend is still present in the market, as consumers may find low advertised fares but often face extra fees, restrictions, and limited flexibility. This approach allows legacy carriers to attract both price-sensitive travelers and those willing to pay more for comfort or convenience, all within the same flight, thus limiting room in the market for LCC.

1.1 Background

While the commercial aviation market has changed, the LCC sector has taken a significant financial hit. This can be seen in the attempted merger between JetBlue and Spirit, United States District Court for the District of Massachusetts (2024). Following the blockage, this Spirit has filed for Chapter 11 bankruptcy twice and faces significant financial problems at the time of this writing, as seen in Grant (2025). The sector problems are not just unique to Spirit. Looking at the two other major players in the LCC sector. Frontier and Southwest have both faced declining profitability and strategic shifts away from their traditional LCC models. These developments indicate that the once powerful force of low-cost competition may be weakening or even disappearing altogether.

This change raises serious concerns for market efficiency, consumer welfare, and the overall structure of the airline industry. If LCCs continue to exit markets or scale back operations, the resulting reduction in competition could enable legacy carriers to obtain even greater pricing power or reduce service quality without fear of their

losing large market share. Higher fares and fewer route options would disproportionately affect lower-income travelers who rely on affordable air travel from LCCs. Moreover, it remains unclear whether the "Basic Economy" products introduced by legacy carriers are perfect substitutes for LCC services. If these products are not perfect substitutes, the erosion of low-cost competition could have a larger economic impact than is yet realized. From a policy standpoint, these changes challenge the measures of competition and consumer choice that regulators such as the Department of Transportation (DOT) and Department of Justice (DOJ) rely on when reviewing mergers and enforcing antitrust laws.

1.2 Empirical Strategy

This paper seeks to address these issues by asking a simple question: What happens to airfares when an LCC exits a U.S. domestic market? While earlier studies have primarily focused on the entry of low-cost carriers and their effects on pricing, this paper takes a different perspective by examining the impact of exit, a topic that has received far less empirical attention. Understanding the effects of LCC withdrawal is critical for determining whether the price discipline traditionally associated with low-cost competition still holds in today's concentrated airline industry. By focusing on exits, this study contributes to the broader literature on market structure and competition, offering new evidence on how industry consolidation and strategic behavior by legacy carriers interact to shape fare outcomes. Methodologically, this paper employs a difference-in-differences (DiD) approach. In Q3 2023, Frontier pulled out of Albany International Airport, creating an exit shock on several nonstop and connecting routes. The model considers routes to be treated if they are directly affected by the exit in Albany. For the control, the model uses similar routes that are in Syracuse. Using a panel data set of U.S. domestic routes, the model estimates the causal effect of LCC exit on average fare using an interacted DiD variable that

combines post and treatment variables. The model controls for variables such as passenger supply and demand, distance, GDP growth, and jet fuel prices. This model was inspired by much of the existing literature on airline entry and exit.

1.3 Paper Structure

The model finds a statistically significant difference between flight prices before and after the exit at the 1% level. The model estimates a coefficient value of 0.069 for the DiD variable. When expressed as a percentage, the value can be interpreted as a 7.15% increase in prices in Albany compared to Syracuse control routes following the exit of Frontier. Based on this result, the policy ensures that inactive laws and regulations intended to protect the LCC business model against unfair competition are enforced.

The paper is structured as follows: Section I reviews the existing literature on airline competition and fare determination, including the effects of deregulation, entry, and pricing strategies. Section II describes the data sources, variable construction, and descriptive statistics. Section III outlines the empirical methodology, detailing the identification strategy and estimation model. Section IV presents the main results and robustness checks, while Section V concludes by summarizing the key findings, discussing their policy relevance, and suggesting avenues for future research.

CHAPTER TWO

Literature Review

To understand how airline competition works today, it is important to understand how the industry has changed over time. The U.S. airline market has gone through major shifts since deregulation. As the market has evolved, airlines have changed how they price tickets and compete to serve different consumer groups. This section examines what previous research has found about airline economics and how it relates to this paper's research question. The paper begins by reviewing the basic economic theory of the airline industry through the lens of competition and consumer behavior. The review then moves on to applying these theories directly to the present-day industry. Over time, mergers, network growth, and consolidation have all played major roles in shaping how the commercial aviation market has become what it is today. The section also explores how consumer behavior trends have shaped the airline industry into a market that comes with high price inelasticity. The review then turns to previous empirical studies that look at a similar research question to this paper and considers how what was learned in those studies can be applied here. In the end, this section aims to show how those insights connect to the current state of airline competition and pricing.

2.1 Economic Theory on Competition in the Airline Industry

The economic theory that supports this paper provides the background for understanding how competition and market structure influence airfare prices. Stigler (1964) is essential for understanding the fundamentals of the aviation market as an oligopoly. The paper explains that in an oligopolistic market, where only a few firms dominate an industry, those firms often act to limit competition and increase their market power. Stigler (1964) argues that because of the opportunity to maximize profits, firms have strong economic incentives to coordinate their behavior with other firms. By doing this, a firm can restrict competition. In the commercial aviation market, coordination is feasible, as ticket prices are listed online. In practical terms, when only a few carriers dominate a route or an airport, they can indirectly align pricing strategies or control capacity. This framework is important to this paper because it helps explain how market concentration can lead to higher prices for consumers, even when competition technically still exists.

Following years of deregulation, the foundation outlined in Stigler (1964) began to appear clearly in the data. Borenstein (1989) shows that when airlines dominate a specific airport, they can charge higher ticket prices. From this hub dominance, airlines can exercise market power and raise prices, consistent with Stigler (1964). The findings were later reinforced in Borenstein and Rose (2014), which documents how regulatory reform, mergers, and consolidation have produced less competitive markets and increasingly concentrated hub structures. These studies show that even though deregulation was originally intended to promote competition, the long-term result has given a concentrated market dominated by a few carriers at hub airports.

This pattern is not limited to the United States. Evidence from international markets shows similar outcomes. In Fu et al. (2010), the authors find that when international markets first opened to more competition, consumers initially benefited

from lower fares. However, these gains were not long term. Airlines began forming global alliances that allowed them to coordinate routes, share capacity, and strengthen their market positions. This essentially created what is known as a legal Cartel. This has reduced competition, resulting in higher fares that hurt consumers. These results directly relate to Stigler (1964) broader idea that markets often tend to move back toward consolidation as firms adapt to competitive pressures.

Together, these studies form the theoretical framework for understanding how pricing power develops in airline markets. They show that reduced competition, whether through regulation or mergers, can and do lead to higher fares and fewer options for consumers. This framework provides an important foundation for this paper's analysis, which examines whether the same principles continue to hold in the modern U.S. market. Along with this, the literature helps motivate the question of this paper: whether the market responds to LCC exit in the same way it did to LCC entry, as earlier studies observed.

2.2 Beyond Price: Service Quality and Brand in Airline Choice

While economic theory helps explain the relationship between market concentration and prices, more recent research shows that consumer behavior has become increasingly important in how airlines compete. Unlike many industries, airline competition is driven by both price and a firm's reputation with consumers. Passengers in the market today are responding differently to shifts in competition than in the past. Earlier studies, such as Morrison (2001), found that when an LCC entered a market or a nearby market, fares dropped and demand increased, implying greater consumer surplus. However, more recent research suggests that consumers now consider other factors when choosing an airline. Factors such as reliability, service quality, brand reputation, and emotional connection with an airline have become more important in shaping demand and willingness to pay, changing how competition works in the

industry.

A useful starting point for understanding these behavioral shifts is Prince and Simon (2015), which examines how incumbent airlines adjust service quality when an LCC, in this case, Southwest, enters a market. The study looks at changes in on-time performance to capture how firms respond to new competition. While it might be expected that airlines would improve service to remain competitive, the results show the opposite. Service quality at legacy carriers actually declines after Southwest's entry, according to this study. The paper by Prince and Simon (2015) argues that this behavior reflects a cost-cutting strategy in which incumbents reduce operational quality to lower costs and compete on price while maintaining profits. This finding challenges the idea that more competition automatically improves service quality. Rather than improving service, carriers often make trade-offs between efficiency and the overall customer experience to increase firm profits

Other research supports this idea by showing that passenger demand has become increasingly inelastic. Ragab et al. (2024) use survey data to show that airline service quality, perceived value, and brand loyalty all significantly influence passengers' willingness to pay. Their results reveal that even when prices increase, travelers often stay loyal to airlines with a better reputation. This gives those airlines an advantage, since they can maintain higher fares without losing customers. As a result, competition in today's aviation markets is not driven solely by air fares but also by passengers' personal views of a firm.

Taken together, these studies show that airline demand has become more price-inelastic and more reliant on brand reputation. This trend also helps explain the success of the "Basic Economy" fares offered by major legacy carriers. These fares allow legacy carriers to appeal to consumers who may be more price-elastic while still maintaining their reputation with higher-paying customers across different fare classes. In this way, airlines can capture both segments of the market without fully

engaging in price wars and sacrificing their premium product.

2.3 Market Structure and Antitrust in the U.S. Airline Industry

Building on the consumer behavior section, another major factor in the market change has been the number of firms operating. Since a shift in public policy toward market regulation, we have seen the oligopoly market grow stronger. The way the industry is moving has raised significant concerns about increased predatory pricing, multi-market contracts, and further consolidation driven by mergers. Each of these problems has allowed the market to reach its current state. As a result, this has led to a decrease in competition driven by LCC's exit from individual markets.

Pinto (2009) provides an early foundation for understanding airline competition under deregulation. The author argues that the industry's high fixed costs create a high barrier to entry. As a result, the industry is vulnerable to predatory pricing practices. Firms do this by matching or undercutting entrants to protect market share even as prices approach marginal cost. The author also points to another advantage these carriers have: the size of the firm. This allows them to have well-run frequent flyer programs. This increases the likelihood that they will be perceived as having a good reputation, which ties back to the consumer behavior theory section. To go along with the ability to keep slots that may cause temporary loss of cash, in order to force out LLCs that do not have the ability to do this due to lower profit margins. As a result, the presence of new entrants, such as LCCs, does not always lead to long-term price pressure. This dynamic helps explain why the impact of LCC exits may be more permanent than earlier literature suggested; thus, once LCCs disappear, dominant carriers can restore higher fares with limited threat of reentry.

Further evidence of the weakening of competition appears in research on multi-market contact. Bilotkach (2011) finds that after the 2005 merger between America West and US Airways, airlines across the industry began competing less aggressively,

consistent with the "mutual forbearance" hypothesis. This occurs when carriers meet in many markets; they are less likely to engage in price or service competition out of fear of retaliation in other markets. The study's DiD analysis shows that greater multimarket contact led to reduced flight frequency, which limits capacity and benefits incumbents but limits consumer choice. These findings show that consolidation can weaken both price and service competition even without explicit collusion, a concern directly relevant to modern U.S. market conditions.

Recent policy and legal developments underscore these same risks. In. In United States District Court for the District of Massachusetts (2024), the District Court for Massachusetts blocked JetBlue's proposed acquisition of Spirit, concluding that the merger would "substantially lessen competition" and harm consumers who rely on Spirit's low-cost model. The court found that Spirit's role in exerting "downward price pressure" was unique and that its removal would make it difficult for other airlines to replicate similar pricing discipline. The decision emphasizes how LCC, even if it does not own the largest market share, still puts downward pressure on prices.

Edelman (2021) extends this discussion to larger scope policy implications, arguing that weak antitrust enforcement and growing common ownership among airlines have further limited competition. Institutional investors now hold significant stakes across multiple airlines. Thus, it creates conditions ripe for coordinated price discrimination among these firms. Edelman (2021) argues for stronger enforcement under Section 411 of the Federal Aviation Act, which grants the Department of Transportation authority to regulate "unfair methods of competition." His argument highlights a key policy gap, while traditional merger reviews address direct consolidation. The litigation fails to address structural issues within the two merging firms.

Together, these studies and cases help show a clear shift in the U.S. airline industry. Consolidation, multimarket overlap, and weakened antitrust enforcement have

reduced the competitiveness that allowed LCC to put downward pressure on prices. For this paper, these findings reinforce the need to test whether LCC entry and exit still meaningfully influence fares in today's market. If modern competition is shaped more by brand loyalty, market concentration, and shared ownership than by price rivalry, then the traditional mechanisms that once kept airfares low may no longer function as they did.

2.4 The Role of LCC Price Competition

LCCs have played an essential role in shaping the current structure and environment of the airline industry. The introduction of a new business model increased market efficiency, making air travel more accessible to consumers. The LCCs have also challenged the way the traditional legacy carriers operate. The rise of these carriers has reshaped competition on U.S. domestic routes. As a result, legacy airlines have had to adapt their pricing and service strategies. However, LCCs are more vulnerable to macroeconomic and policy changes, as well as other shifts in the economy, than traditional legacy carriers. Understanding how these factors interact helps explain not only historical price changes but also how the current market may respond to LCCs' withdrawals or financial struggles.

2.4.1 Business Model and Policy Environment of LCCs

LCCs have reshaped the economics of the United States airline industry by prioritizing affordability for consumers. Their influence has been especially significant in smaller or mid-sized markets where major airlines face limited competition. Bachwich and Wittman (2017) examine the rise of the LCC model in the United States and show that carriers such as Spirit, Frontier, and Allegiant operate with a business strategy centered on high aircraft utilization, simple fare structures, and additional revenue from unbundled add-on services. The authors find that ultra-LCC presence

reduces average base fares by approximately 21% compared to markets without them, while traditional LCCs reduce fares by around eight percent. This difference suggests that the ultra-low cost model plays a particularly powerful role in promoting price competition across U.S. domestic markets. However, Bachwich and Wittman (2017) also note that ultra-LCCs tend to exit markets more frequently, abandoning roughly 26% of new markets within 2 years of entry. This volatility demonstrates that the benefits they bring to consumers can be temporary when market conditions or policy environments are unfavorable.

The authors further explain that ultra-LCCs differ fundamentally from traditional LCCs such as Southwest and JetBlue. While those airlines have gradually moved toward hybrid models that blend low fares with some service amenities, ultra-LCCs have maintained a true-only-seat business model, with everything else sold as an additional purchase. This approach, while effective in generating low fares, leaves these carriers more exposed to financial risk. Because they rely on ancillary revenue and operate with tight margins, shocks such as rising fuel prices or labor disputes can severely affect their profitability more than the traditional LCCs. Thus, it shows that it is important to differentiate between LCCs and Ultra LCCs

The current policy environment also plays a crucial role in determining how these airlines perform. Abate et al. (2020) studied how government support programs implemented during the COVID-19 pandemic affected airline competition. Their research finds that financial assistance during the crisis was distributed unevenly, favoring larger network carriers that had greater overall influence. These airlines received proportionally larger bailouts and recovered faster. This builds on the fact that this paper has already explained that LCCs are more affected by shifts in economic conditions. This limited LCC's ability to maintain service in smaller and medium-sized markets. The findings of Abate et al. (2020) suggest that public policy can unintentionally undermine market competition if it provides uneven support to larger carriers. Taken

together, these studies demonstrate that the structure and survival of the low-cost sector depend on both internal and external decisions arising from economic shocks.

2.4.2 Empirical Evidence on Entry and Fare Effects

Empirical research has consistently shown that LCC entry generates strong downward pressure on airfares. Morrison (2001) provides one of the foundational studies on this topic, analyzing how Southwest Airlines' expansion affected U.S. markets. The study concludes that both actual and potential competition from Southwest significantly reduced fares. The paper estimates that these fare reductions saved consumers nearly 13 billion dollars annually in 1998. The analysis also establishes a framework that continues to inform modern competition studies in the airline industry.

Building on this work, Goolsbee and Syverson (2008) explore how incumbent carriers respond to the threat of entry from Southwest Airlines. Their results show that even the expectation of future competition can lead incumbents to lower prices. This finding reveals that the threat of LCC competition can have almost as strong an effect as actual entry, suggesting that perceived market competition can help maintain competitive prices even without actual entry into the market.

Further evidence comes from studies examining different types of responses across market structures. Ito and Lee (2003) find that hub-dominated carriers generally respond moderately to the entry of low-cost competitors. They tend to reduce fares slightly but avoid aggressive price wars that could harm profitability. This indicates that competitive responses vary depending on the level of market concentration and the importance of the route in a carrier's network. Similarly, Tan (2016) finds that when LCCs enter new routes, legacy airlines reduce both average and premium fares, thereby narrowing the price gap between business and leisure travelers. These effects demonstrate how low-cost entry enhances consumer welfare beyond simple fare reductions.

Other studies highlight the geographic dimensions of competition. Brueckner et al. (2013) show that LCCs can influence fares not only on the routes they directly serve but also in nearby markets, such as adjacent airports within the same metropolitan area. Their research finds that fares in those markets decline significantly when an LCC operates nearby, suggesting that consumers are willing to substitute between airports to take advantage of lower prices. This spatial competition amplifies the broader market impact of LCC presence and reveals how competition extends beyond specific route pairs.

The most recent comprehensive evaluations, such as Brueckner et al. (2013) and Bachwich and Wittman (2017), collectively demonstrate that low-cost and ultra-LCCs continue to play a defining role in fare competition. However, they also indicate that the magnitude of this effect may have weakened over time as legacy carriers have adopted elements of the low-cost model through products such as Basic Economy fares. These hybrid strategies allow major airlines to retain price-sensitive travelers while preserving brand loyalty and service differentiation for higher-paying customers. As a result, the clear divide that once existed between legacy and LCCs has begun to blur, raising important questions about whether LCC exits will now have the same impact on prices as they did in earlier decades.

In addition to providing empirical evidence to support the competitive effect of an LCC on the market, these papers also provide the empirical design for a model. As seen in Morrison (2001), the author uses airport-pair-level data to conduct their study of the Southwest effect reported in the paper. Goolsbee and Syverson (2008) use an event study with fixed effects to capture the effect of entry. The event study framework, when combined with the same fixed used in this paper, is extremely similar to the DiD equation employed in this paper's empirical section.

Overall, the evidence from past studies shows that low-cost carriers and ultra-LCCs have historically reduced fares and improved accessibility across markets, while

their withdrawal tends to reverse those benefits. The persistence of these findings across different time periods, market structures, and carrier types provides strong support for studying how recent exits, such as Frontier's withdrawal from Albany International Airport, may influence competition and consumer affordability today.

2.5 Contribution to the Area of Research

This paper shifts the focus of recent airline economics literature from LCC entry to LCC exit in the market. Almost all prior research in this area has focused on the entry of LCC carriers, which many papers refer to as the Southwest effect. These studies consistently show that the entry, or even the threat of entry, of an LCC like Southwest can and does put downward pressure on ticket prices. However, little research has been done on the exit effect of LCC, creating a gap in the literature. This gap in the literature is particularly relevant in the current post-consolidation airline industry, where market concentration is higher, and several low-cost carriers have faced financial difficulties in recent years. By applying the same empirical methods used in prior research, this paper aims to address that gap

CHAPTER THREE

Data

This paper collects publicly available data from the Department of Transportation (DOT) and the Bureau of Transportation Statistics (BTS). Specifically, the paper uses the DB1B Market and T100 market data. This data is used in a large portion of the aviation economics literature, including Morrison (2001), Ito and Lee (2003), and Tan (2016), among others. The data is a sample of tickets sold by U.S. carriers on domestic and international routes; specifically, the data is self-reported at a 10% rate. The paper uses data from Q4 2021 to Q4 2023. This allows the paper to cover roughly 1 year pre-event and 1 year post-event.

The data is observed at the granularity of (i, j, t), where t is defined as the quarter, j as the carrier, and i as a route with the same origin and destination airports. This paper does not account for connections in the granularity of the data set. Thus, $(ALB \to JFK \to LAX)$ is the same as $(ALB \to ORD \to LAX)$ on the route level. Doing this allows the paper to aggregate at this level and calculate the mean fare, as well as other variables such as passenger counts and average distance.

The treatment group consists of routes originating from or arriving in Albany that were directly affected by Frontier's withdrawal, while the control group includes structurally similar routes in Syracuse that retained Frontier service during the same period. This setup ensures comparable airport size, geographic location, and passenger demographics, making it an optimal choice for the control group

The full set of variables is presented in Table 1.1 in the appendix. Data cleaning

and feature engineering steps include merging quarterly DB1B files with the T100 dataset, using different features to engineer new features, and computing averages across markets. The resulting nine quarters of data yielded approximately nine quarter observations.

The summary statistics are presented in Table 3.1, split by control and treatment groups. In total, the cleaned data set contains 182 observations, with a 104/74 split for control and treatment, respectively. The paper notes that this relatively small data set is a limitation for the result. Overall, the summary statistics suggest that the control and treatment groups are generally comparable, as the paper does in the DiD methodology. The average logged fare value is similar: 5.11 and 5.18, respectively. The route distances for both groups are also almost identical, with both coming within one mile of 3268.

In terms of dating, cleaning the paper notes the following. While some papers in this academic literature space include prices only in 10%-90% intervals to weed out outliers, this paper includes them due to the limited data sample. Additionally, while the paper does not include fares purchased in bulk (i.e., group purchases), it does include all fare classes. The paper finds this to be reasonable because the markets being looked at do not include wide-body aircraft that would normally come with extensive business and first class.

Table 3.1: Summary Statistics

Panel A: Full Sample

Variable	N	Mean	SD	Min	Max
Log Average Fare	182	5.136	0.406	3.886	6.197
Treatment Route	182	0.429	0.496	0.000	1.000
Post Period	182	0.544	0.499	0.000	1.000
Treatment × Post (DiD)	182	0.203	0.404	0.000	1.000
Log Passengers	182	7.866	2.228	0.000	10.080
Seat Capacity	182	6874.187	5644.717	0.000	23850
Route Distance (miles)	182	3267.896	946.273	1053	4830
GDP Growth	182	0.019	0.007	0.012	0.035
Jet Fuel Price	182	318.693	56.372	233.440	420.601

Panel B: Treatment Group

Variable	N	Mean	SD	Min	Max
Log Average Fare	78	5.110	0.393	3.886	5.927
Treatment Route	78	1.000	0.000	1.000	1.000
Post Period	78	0.474	0.503	0.000	1.000
Treatment × Post (DiD)	78	0.474	0.503	0.000	1.000
Log Passengers	78	7.899	2.557	0.000	10.080
Seat Capacity	78	8084.295	6631.214	0.000	23850
Route Distance (miles)	78	3267.590	979.299	1073	4830
GDP Growth	78	0.020	0.007	0.012	0.035
Jet Fuel Price	78	320.496	59.745	233.440	420.601

Panel C: Control Group

Variable	N	Mean	SD	Min	Max
Log Average Fare	104	5.156	0.416	3.997	6.197
Treatment Route	104	0.000	0.000	0.000	0.000
Post Period	104	0.596	0.493	0.000	1.000
Treatment × Post (DiD)	104	0.000	0.000	0.000	0.000
Log Passengers	104	7.842	1.958	0.000	9.563
Seat Capacity	104	5966.606	4601.990	0.000	14230
Route Distance (miles)	104	3268.125	925.518	1053	4476
GDP Growth	104	0.019	0.006	0.012	0.035
Jet Fuel Price	104	317.341	53.959	233.440	420.601

CHAPTER FOUR

Methodology

To address the question of whether the exit of an LCC influences airfare prices on U.S. domestic routes, the paper utilizes a DiD framework to estimate the causal effect of a carrier's exit by comparing price changes before and after the event between affected and unaffected markets. In this context, the treatment event is defined as the exit of a low-cost or ultra-LCC from a market where at least one legacy carrier continues to operate. Previous studies examining LCC entries, such as Goolsbee and Syverson (2008), have successfully applied DiD or related interaction-term approaches to estimate competitive responses. Although these studies focus on market entry, the same logic can be extended to analyze market exit, as both events represent structural changes in competition that have a clear event period.

This paper examines Frontier Airlines' withdrawal from the Albany International Airport market in late 2022. Frontier, an ultra-LCC, exited Albany in 2022Q4, as reported by Karlin (2022). As a control group, this paper uses Syracuse Hancock International Airport, which continues to maintain Frontier service, including flights to Orlando and Denver, two routes that Albany lost. Syracuse shares many structural similarities with Albany, including geographic proximity in upstate New York, comparable population sizes, and similar airport capacities, making it a strong control market.

To validate the use of this methodology, this paper conducts parallel trends tests, confirming that the DiD framework is appropriate for our empirical setting. This

paper also tests for potential spillover effects between Albany and Syracuse, given their proximity of roughly 2.5 hours by car. As shown in Figures 4.1 , there is no noticeable increase in Syracuse's passenger volumes relative to total market trends, indicating no significant substitution effect between the two markets. Similarly, this paper observes no sharp decline in the number of Frontier-operated flights from Syracuse following the Albany exit. The robustness checks are discussed further in the robustness section of this paper.

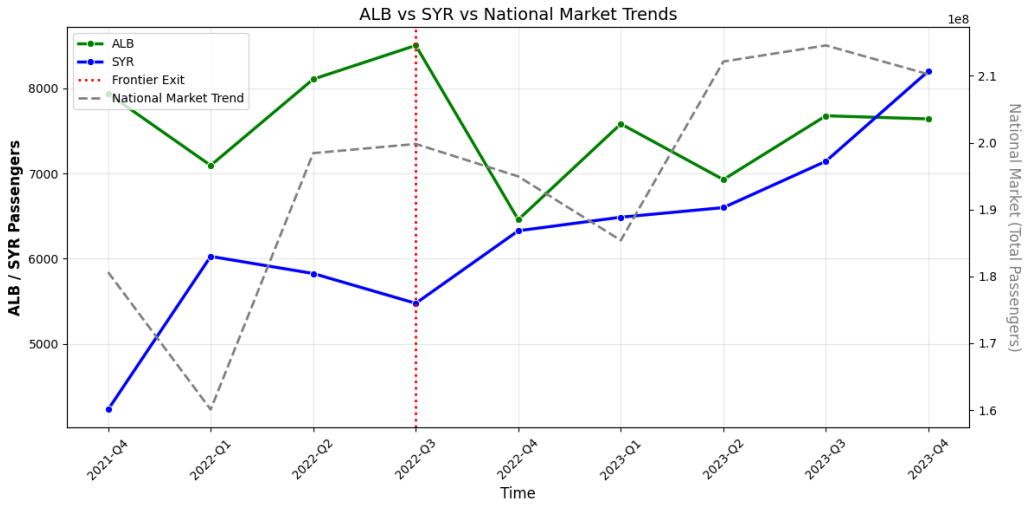


Figure 4.1: Demand Trend National, SYR and ALB

To capture changes in fare levels, this paper uses the natural logarithm of the mean fare per route per quarter as our dependent variable. Consistent with prior research, including Ito and Lee (2003), Goolsbee and Syverson (2008), and Morrison (2001), this log transformation allows coefficients to be interpreted as approximate percentage changes and reduces the influence of outliers. The analysis is conducted at the route, carrier, quarter (i, j, t) level of granularity. The dependent variable is calculated as follows:

$$\ln\left(\frac{\sum FareValue_{ijt}}{NumberOfFares_{ijt}}\right)$$

The paper's key independent variable is an interaction term that captures the causal effect of an LCC's exit on fares. This interaction term combines two binary

variables. The first, $Post_t$, returns 1 for periods after the exit (i.e., Q4 2022 onward) and 0 otherwise. The second, $Treatment_{ij}$, returns 1 for airport pairs affected by the exit, i.e., routes that contain Albany International as a point of origin or destination, and 0 for the control group, i.e., routes that contain Syracuse as a point of origin or destination. Together, the interaction term ($Treatment_{ij} \times Post_t$) isolates the effect of Frontier's withdrawal on fare levels. The corresponding coefficient,β_1 , measures how much more or less the price in the treatment group changed, after accounting for the control group.

A positive and statistically significant β_1 would indicate that Frontier's exit led to higher average fares, consistent with reduced price competition. Conversely, an insignificant or negative β_1 would suggest that competitive pressure remained strong, perhaps due to substitution by legacy carriers' Basic Economy fares or other low-cost alternatives that remain in the market.

This paper includes a comprehensive set of control variables to account for market characteristics that may also influence prices. Starting with route distance, the longer the route, the higher the operating cost, thus resulting in higher fare prices. Second, controlling for the number of seats available on a flight is a direct proxy for supply in a given market. A lowering in supply can cause higher prices. Finally, on the other hand, passengers on a flight serve as the proxy for demand in the model. The higher the demand, the higher the carrier can charge for ticket prices. The model also has two macro control variables. First, GDP growth helps stabilize overall economic conditions. While volatile fuel prices are a significant input cost for operating flights, including them as a control for ticket prices is crucial.

In addition to control variables, this paper also includes. route, carrier, and time fixed effects to control for unobserved heterogeneity. Starting with route-fixed effects, it is important to examine whether certain routes exhibit time-invariant characteristics. Carrier fixed effects controls for unique carrier characteristics, such as business

model, brand reputation, and other specific traits. Finally, time-series effects capture seasonal fluctuations in the market. such as persistent route characteristics and seasonal demand fluctuations. The full econometric model is presented below with fully written-out control variables:

$$\begin{aligned}\ln(Fare_{ijt}) = \beta_0 + \beta_1(Treatment_{ij} \times Post_t) + \beta_2 \ln(DemandPassengers_{ijt}) \\ + \beta_3 SupplyPassengers_{ijt} + \beta_4 Distance_{ij} + \beta_5 GDPGrowth_t \\ + \beta_6 JetFuelPrice_t + \mu_i + \tau_t + \gamma_j + \varepsilon_{ijt}\end{aligned}$$

Here, μ_i represents route fixed effects, τ_t denotes time fixed effects, γ_j captures carrier fixed effects, and ε_{ijt} is the error term. The empirical model will also have standard errors clustered at the route level to account for potential within-route correlation over time.

CHAPTER FIVE

Empirical Results

This paper presents the results of the regression analysis in Table 5.1. This table shows the gradual build-up to the full model, with Column 1 showing the regression results as a baseline with only the DiD coefficient displayed and no fixed effects. Columns 2 through 4 each add a new fixed effect, going with route-time-carrier fixed effects in that. Lastly, columns 5-9 introduce new control variables, leading to the full model being deployed in column 9. Across all models, the DiD coefficient remains positive and statistically significant, as desired.

In the full model column (9), the did coefficient is statistically significant at the 1% level, with an estimated effect of a 7% increase in average fares following the exit compared to control routes. It should be noted that the significance level t and the sign of the coefficient remain stable throughout the model-building process. Additionally, the coefficient strength and the overall model strength increase progressively throughout the model's build. Stability across the specifications increases confidence in robustness.

The fully deployed model was able to capture a large amount of variation in fares, with the final R-squared value coming in at .90. Most notably the inclusion of carrier fixed effects showed the largest increase in R-squared value jumping from roughly .30 to .90. This suggests that carrier specific characteristics that are not included in control variables count for a large variation in fare value. In particular, carrier fixed effects absorb carrier-specific differences across airlines, such as average

price levels, brand reputation, and firm cost structures. This is consistent with the paper's hypothesis that low-cost carriers place downward pressure on the average price of a ticket in the market. In the fully deployed model, all control variables behave as expected: markets with higher demand are associated with lower fare prices, underscoring the importance of highly competitive markets in driving prices down. Additionally, the two macro control variables are statistically significant, suggesting that broader economic conditions also influence fare outcomes.

Table 5.1: Effect of Frontier Exit on Log Average Fares

	Dependent Variable: ln(Average Fare)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treatment × Post	0.251***	0.243***	0.276***	0.076***	0.072***	0.062***	0.069***	0.069***	0.069***
	(0.040)	(0.042)	(0.044)	(0.019)	(0.019)	(0.017)	(0.020)	(0.020)	(0.020)
Post Period	-0.090***								
	(0.031)								
Treatment Market	-0.176	-0.168							
	(0.173)	(0.176)							
log(Passenger Demand)					0.011*	-0.003	-0.022**	-0.022**	-0.022**
					(0.006)	(0.006)	(0.009)	(0.009)	(0.009)
Route Distance							0.000	0.000	0.000
							(0.000)	(0.000)	(0.000)
GDP Growth Rate								0.163***	0.200***
								(0.003)	(0.003)
Jet Fuel Price									0.010***
									(0.000)
Constant	5.210***	5.079***	5.095***	5.365***	5.291***	5.370***	5.340***	5.335***	3.076***
	(0.136)	(0.176)	(0.050)	(0.023)	(0.065)	(0.051)	(0.061)	(0.061)	(0.039)
Route Fixed Effects	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Carrier Fixed Effects	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Time Fixed Effects	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	182	182	182	182	182	182	182	182	182
R^2	0.027	0.057	0.301	0.896	0.898	0.901	0.905	0.905	0.905

Overall, these findings align with the majority of the existing literature on general commercial airline competition. While most prior research focuses on LCC entry rather than exit, the magnitude and statistical significance of the estimated DiD coefficient align with earlier evidence that LCC introduces competition, exerting downward pressure on fares. In this specific case study, the removal of the LCC introduces upward pressure on fares in the affected market.

5.1 Policy Implications

The findings of this paper, when interpreted, can provide several implications for the future of airline competition. First, the exit of an LCC from a city pair route market can put downward pressure on fares. When Frontier exited the market, fares increased by 7%. This suggests that the presence of LCCs constrains the pricing behavior of incumbent airlines. Thus, regulators should pay close attention to ensure LCC remains in the markets.

Regulators can do that by doing the following: first, preventing mergers that would eliminate a LCC business model, as they did in the JetBlue Spirit attempted merger. Additionally, regulators such as the FAA should look into protecting landing slots and gates at airports for LCCs. By doing this, they can help airlines avoid being pushed out and maintain higher airport capacity, helping keep down their costs and increase operational efficiency. In times of economic hardship, Congress must ensure stimulus is distributed fairly and equally, unlike during COVID. They should also continually be exploring new policies in the evolving market to keep LCC competitive.

5.2 Robustness Check

To confirm that the model's estimated effect of Frontier's exit from the Albany market is driven by that event, this paper conducts several robustness checks that examine model specification assumptions, trends, and demand shocks arising from the exit of a market. To be specific, this paper checks the classic DiD parallel trends assumption, event study specification, airport demand trends at the local and national level, and finally, the paper removes the event period from the data set to take the noise seen in the parallel trend graph out of the model to see if results hold.

One of the most important assumptions in a DID model study is the parallel trend study. This shows that the control and treatment group's dependent variables,

in this case the logged average fare for a route, are parallel leading up to the event time period. Upon reaching the even time point, the assumption expects the treated group to deviate from its original path. As seen in Figure 5.1, this is exactly what the graph shows: a major dip, followed by a spike in prices in the post-event period for the treatment group. Also important to note is that there is no movement in the time since the last pre-treatment period, showing there is no pre-planning price drop by other airlines in the market.

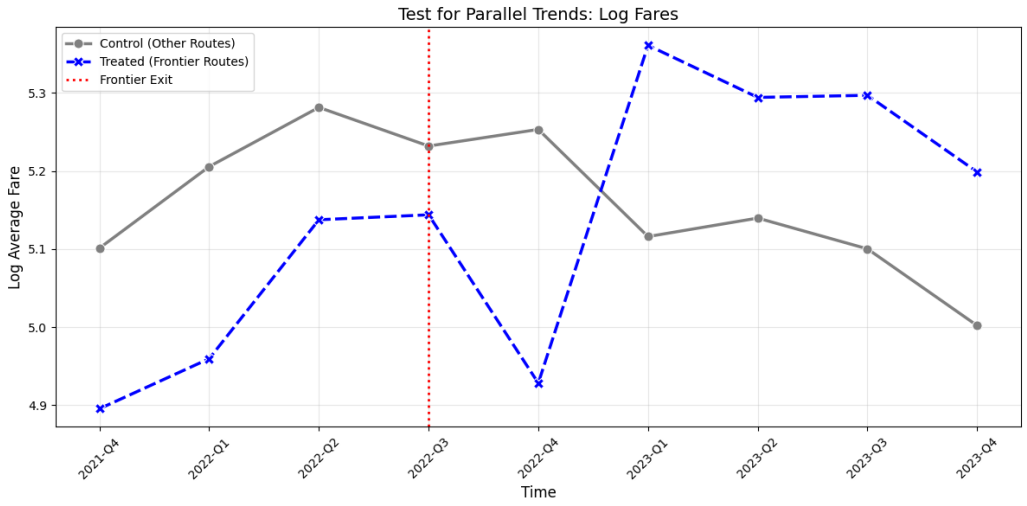


Figure 5.1: Parallel Trend Test Control Vs Treatment

The paper also conducts further testing on the chosen treatment and control groups, using an event study with a separate DiD model than the one used to estimate the causal effect on logged airfare. This model estimates the effect for each period pre- and post-exit. This helps confirm the parallel-trend test assumption discussed in the previous paragraph. The goal of this study is to show no statistically significant difference in zero in the pre-exit periods and statistically significant differences from zero in the post-exit periods. The results of this study are shown in Figure 5.2 and Table 5.2. The graph and table indicate that the assumption holds: there is no significant difference in the pre-event period. This shows that the parallel trend assumption holds there. However, the paper finds no significant difference in the first

post-event period, and interprets this as evidence that it took time for firms to adjust prices. That conclusion is derived from the following two reasons. One, it is well known that airfare tickets can be purchased months in advance. In this data set, regardless of the purchase date, the ticket price is recorded in the quarter and year of departure, so a significant number of tickets may have been purchased pre-event. Secondly, and more importantly, the following time points are significantly different from zero, indicating that the control and treatment groups diverge. However, it must be noted that the pre-treatment analysis includes only three pre-treatment quarters, providing a limited sample to establish the parallel trends assumption.

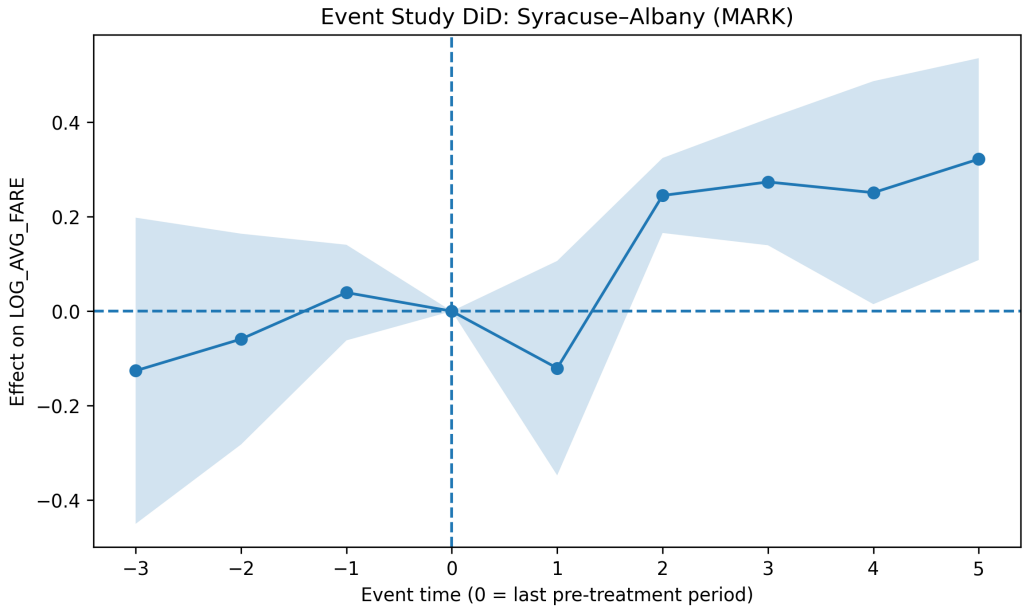


Figure 5.2: Event Study Test Control Vs Treatment

Table 5.2: Event Study: Effect of Frontier Exit on Log Average Fares

	Coefficient	Std. Error	t-stat
Event $t-3$	-0.1261	(0.1653)	-0.76
Event $t-2$	-0.0591	(0.1138)	-0.52
Event $t-1$	0.0394	(0.0516)	0.76
Event $t+1$	-0.1205	(0.1158)	-1.04
Event $t+2$	0.2449***	(0.0405)	6.05
Event $t+3$	0.2735***	(0.0684)	4.00
Event $t+4$	0.2508**	(0.1204)	2.08
Event $t+5$	0.3220***	(0.1090)	2.95
log(Passenger Demand)	-0.0927***	(0.0277)	-3.35
Passenger Capacity	3.47e-05***	(1.29e-05)	2.70
Route Distance	5.90e-05	(6.69e-05)	0.88
Observations		182	
R-squared (Within)		0.142	
Entity FE		Yes	
Time FE		Yes	

Notes: The dependent variable is log(Average Fare). Event-time indicators are relative to the quarter prior to Frontier's exit. Standard errors clustered at the route level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

After examining tests related to the parallel trends assumption, the paper turns to the spill-over effect. This test is essential for this study due to the proximity of the Albany and Syracuse airports, which are 2 hours apart. Specifically, the paper is curious to see whether traffic changes from Albany to Syracuse in the post-event period. To examine this, the paper creates the graph seen in Figure 4.1. From this, it

can be seen that the following two things are true. Both airports seem to experience the same seasonal demand cycles. Secondly, the control airport, Syracuse, does not experience a major spike in its demand in the post-event time period. An important note is that demand at Albany drops during the event period. The paper notes that this is expected due to the loss of an entire carrier (Frontier) during this period.

Lastly, the paper uses a robustness check to address the noise in the data seen in some of the previous checks around the event period. To do this, the paper runs the exact same DiD estimation model seen in the methodology section. The only difference is that the paper drops the event period from the data, expecting the magnitude and sign of the DiD coefficient to remain unchanged. As seen in Table 5.3, this is exactly what the results show. This shows that the treatment effect on the post-treatment coefficient is preserved even in the presence of the noise introduced by the event period.

Table 5.3: Estimates of Frontier's Exit on Log Average Fares Drop Event Period

	Dependent Variable: ln(Average Fare)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Route Distance							0.000*	0.000*	0.000*
							(0.000)	(0.000)	(0.000)
Intercept	5.210***	5.079***	5.119***	5.398***	5.364***	5.435***	5.408***	5.402***	2.951***
	(0.136)	(0.176)	(0.047)	(0.034)	(0.053)	(0.026)	(0.034)	(0.034)	(0.041)
log(Passenger Demand)					0.005*	-0.007***	-0.021***	-0.021***	-0.021***
					(0.003)	(0.003)	(0.008)	(0.008)	(0.008)
Post Period	-0.122***								
	(0.040)								
Passenger Capacity						0.000**	0.000**	0.000**	0.000**
						(0.000)	(0.000)	(0.000)	(0.000)
Treatment Market	-0.176	-0.168							
	(0.173)	(0.176)							
Treatment × Post	0.382***	0.368***	0.400***	0.149***	0.145***	0.131***	0.132***	0.132***	0.132***
	(0.043)	(0.046)	(0.050)	(0.017)	(0.016)	(0.015)	(0.015)	(0.015)	(0.015)
GDP Growth Rate								0.171***	0.190***
								(0.003)	(0.003)
Jet Fuel Price									0.010***
									(0.000)
Route Fixed Effects	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Carrier Fixed Effects	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Time Fixed Effects	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	160	160	160	160	160	160	160	160	160
R^2	0.060	0.094	0.367	0.902	0.902	0.905	0.908	0.908	0.908
Adjusted R^2	0.042	0.039	0.301	0.887	0.887	0.889	0.891	0.891	0.891

Through a thorough group of robustness checks, the paper confirms the use of the data and methodology chosen. The group of robustness checks does the following: (i)

checks that there is no evidence of difference in pre-treatment trends between control and treatment group; (ii) there is no demand shock seen resulting in a spillover effect. (iii) No difference in the national macro effects that explain the causal effect on DiD variables. (iv) Show the results are not driven by noise seen in the event period. When combining these four findings together, the results provide suggestive evidence that the interpretation that Frontiers exists has a statistically significant effect on airfare prices, which is the leading factor shown by the chosen model and data

5.3 Limitations

The findings of this paper are limited in a few ways. First, it's noted that the pre-treatment period is only 3 quarters long. Specifically, this limits the validity of the parallel trends test conducted in the robustness checks section. While the DB1B dataset is the industry standard due to the paper's short pre-treatment period and the small airport, the 182 observations constitute a limited sample, reducing the power of the results and introducing unwanted noise to the dataset.

The paper also notes that the empirical test was conducted in only one geographic setting. From this, the results may have limited external validity. The paper focuses on Frontier's exit from routes at Albany International Airport and compares it with Syracuse, a control market. While this setting provides a natural experiment, the estimated effects may differ in larger markets or in routes with different competitive structures than these two smaller markets. The paper notes that the results could differ in a market with multiple LCCs operating.

CHAPTER SIX

Conclusion

The paper asks a simple question: what happens to airfare prices when a low-cost carrier leaves a U.S. domestic market where a legacy carrier is still engaged? To measure this, the paper uses Frontier's exit from the Albany market in late 2022 as a case study of a low-cost carrier's exit. The analysis provides direct quantitative evidence that low-cost carriers in the U.S. domestic market still matter in the new-day U.S. commercial airline market.

For this, the DiD model shows that the frontier exit leads to a statistically significant increase in fares of $\tilde{7}\%$ across affected routes. The model employs route and time fixed effects, carrier fixed effects, controls for passenger volume, and multiple macroeconomic controls to reach this result. The large increase in explanatory power after adding carrier fixed effects indicates that airline-specific characteristics, such as brand reputation, network structure, and service quality, account for a meaningful share of fare variation. As seen above, the paper conducts several robustness checks that confirm the use of the data and methodology to obtain these results.

By reviewing literature from previous eras, the paper adds to an extensive body of work by developing a framework for change in the general market structure. Prior to the writing of this paper, much of the existing literature focused on general entry by low-cost carriers, as they thrived in the early 2000s and 2010s. This paper looks at the opposite, as low-cost carriers have struggled. The market has seen some exits in the early 2020s, with more pullback announcements from Frontier and Spirit in

2024 and 2025. Thus, this paper provides a recent case study to provide evidence on the effect in the post-COVID time period

However, there are limitations to what readers can conclude from this paper. To start, this is one exit of one firm focused on one smaller geographic setting. The paper also shows that a single control airport further limits geographic power. The data span a short time period, limiting the number of observations available to estimate the causal effect. Along with this, the data collected for this study focuses on just airfare prices. As learned from the Literature review, an important factor for consumers is also the quality of service; this paper does not measure this in any way. Even with these limitations, the results suggest. The removal of a low-cost carrier from a market produces a long-term, meaningful shift in airfare prices. Future research should build on the foundation laid in this paper. Expanding the geographic settings to include multiple regions simultaneously. Including more than just airfare in a measure of the causal effect, such as product quality and capacity shifts.

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CHAPTER A

Appendix

Table 1.1: Variable Definitions

Variable	**Description**
Year	Calendar year of observation
Quarter	Calendar quarter (1–4)
Origin Airport	Origin airport code
Destination Airport	Destination airport code
Carrier	Reporting airline carrier
Passenger Demand	Total passenger demand on route
Route Revenue	Total passenger revenue on route
Flight Segments	Number of flight coupons (market segments)
Average Fare	Mean ticket fare on route
Log Average Fare	Natural log of average fare
Log Passenger Demand	Natural log of passenger demand
Low-Cost Carrier Indicator	Indicator for low-cost carrier (LCC)
Passenger Capacity	Total passenger seats supplied
Freight Volume	Freight transported on route
Mail Volume	Mail transported on route
Route Distance	Nonstop route distance (miles)
Treatment	Indicator for treated market
Post	Indicator for post-exit period
Treatment × Post	Difference-in-differences interaction term
Year Fixed Effects (Demand)	Year fixed effects in demand dataset
Quarter Fixed Effects (Demand)	Quarter fixed effects in demand dataset
GDP Growth	Quarterly GDP growth rate
Year Fixed Effects (Supply)	Year fixed effects in supply dataset
Quarter Fixed Effects (Supply)	Quarter fixed effects in supply dataset
Jet Fuel Price	Average quarterly jet fuel price